

Sornambal P B.Tech Information Technology

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🔗 Sornambal 🏠 sornambalp 📁 Portfolio

Professional Summary

Agentic AI and Generative AI-focused B.Tech Information Technology student with hands-on experience building Multi-Agent Systems, Retrieval-Augmented Generation (RAG) Pipelines, LLM-Powered Automation Workflows, and Production-Ready AI Applications using Python, LangChain, LangGraph, FastAPI, and modern AI frameworks. Skilled in LangGraph Orchestration, Prompt Engineering, Semantic Search, Vector Databases, Workflow Automation, API Integrations, and Backend AI Development. Experienced in designing and deploying End-to-End AI Systems involving Agent Coordination, Structured Reasoning Workflows, Tool Calling, Function Calling, Semantic Retrieval, and Intelligent Process Automation. Passionate about leveraging Generative AI, Large Language Models (LLMs), AI Agents, and Advanced Machine Learning Techniques to solve real-world business problems through Scalable, Practical, and Production-Oriented AI Engineering Solutions.

Technical Skills

Technical Skills

Programming Languages

Python, SQL, Java, JavaScript

AI/ML & Deep Learning

Scikit-Learn, TensorFlow, PyTorch, Machine Learning, Deep Learning, NLP, CNNs, Feature Engineering, Model Evaluation, Classification, Regression

Generative AI & LLM Systems

LangChain, LangGraph, RAG, AI Agents, Prompt Engineering, LLaMA, FLAN-T5, Structured Outputs, Semantic Search, Autonomous Workflows

Vector Databases & Retrieval

FAISS, Pinecone, Sentence Transformers, Embeddings, Vector Search, Qdrant (Conceptual Familiarity)

Backend & API Development

FastAPI, Flask, REST APIs, API Integration, JSON Processing

Databases & Cloud

MongoDB, MySQL, Hugging Face Spaces, Render, Netlify, AWS (Basic), Azure (Basic)

MLOps & Developer Tools

Git, GitHub, Jupyter Notebook, Docker (Basic), Databricks, PySpark (Learning)

Core Concepts

LLM Applications, Multi-Agent Systems, Workflow Automation, AI Application Development, Problem Solving

Internships

Agentic AI Intern, S Combinator

02/2026 – 05/2026

Remote

- Designed and implemented **multi-agent AI systems** using LangGraph to automate structured business workflows and internal process execution.
- Built prompt-driven agent pipelines with **tool usage, memory, function calling, routing logic, and API integrations** for end-to-end workflow automation.
- Improved workflow reliability through **prompt refinement, validation logic, testing frameworks, fallback handling, and production-oriented orchestration.**

AI/ML Intern, CloudAI Labs

12/2025 – 01/2026

Remote

- Built Python-based ML pipelines for **data preprocessing, feature engineering, and model experimentation** on structured datasets.
- Applied supervised learning techniques and improved model evaluation performance.
- Collaborated on experimentation workflows and improved code quality/documentation.

Data Science Intern, Cognifyz Technologies

06/2025 – 07/2025

Remote

- Performed exploratory and geospatial data analysis** to identify restaurant demand zones and competitive hotspots.
- Built and optimized regression models** for rating prediction, achieving ~10% performance improvement through feature engineering.
- Delivered data-driven insights** and recommendations through visual analysis and team collaboration Repo link [🔗](#)

Data Science Intern, SkillCraft Technology	04/2025 – 05/2025
• Trained and evaluated 5+ supervised ML models, including Random Forest and Decision Tree, on real-world datasets. • Improved classification performance to 81% accuracy using feature engineering and hyperparameter optimization techniques. • Developed an interpretable Decision Tree model to identify key accident-causing factors, achieving 78% accuracy and improved model robustness Repo link	Remote

Projects

Job Application Intelligence System(Agentic AI)
Tech: Python, LangGraph, LLaMA-3, Flask, Gmail API, Twilio
• Built a multi-agent AI workflow system for recruiter email parsing, task tracking, and application workflow automation. • Implemented prompt-based routing, classification, extraction, and decision workflows using LLMs. • Integrated Gmail API, real-time notifications, validation logic, and monitoring dashboard to improve workflow automation accuracy. Git Link
Smart News Summarizer Agent
Tech Stack: Python, FLAN-T5, LangChain, Groq API, Tavily API, Flask
• Built an autonomous LLM-based agent to search, filter, and summarize relevant news articles using a multi-stage generation pipeline. • Orchestrated external APIs and LLM calls to ensure accurate content retrieval and structured summarization. • Deployed the solution on Hugging Face Spaces, providing an interactive web interface for end users. Git Link
AI Notes Chatbot (RAG-Based Question Answering System)
Tech: Python, LangChain, FAISS, SentenceTransformers, FLAN-T5
• Developed a document intelligence system for extracting and answering queries from unstructured documents • Implemented semantic search using embeddings and FAISS vector retrieval • Built document ingestion pipelines for PDFs, DOCX, and TXT formats • Designed workflows to extract structured information from documents using LLM-based processing • Improved response accuracy using prompt optimization and retrieval tuning Git Link
CareCircle - AI Elderly Care Assistant(Full Stack)
Tech Stack: Node.js, Express, MongoDB, React, Socket.IO, Twilio, Groq API
• Developed a full-stack application for medicine management, emergency SOS alerts, and real-time caregiver dashboards. • Integrated secure authentication, live notifications, and SMS/voice-based emergency alert systems. Git Link

Education

B.Tech - Information Technology, KGISL Institute of Technology	2023 – 2027
CGPA: 9.29	Coimbatore
HSC - Bio-Maths, Sri Ragavendra MHSS	2023
Percentage: 81%	Sathyamangalam

Achievements & Recognition

Awarded Academic Topper (B Tech Information Technology) at KGISL Institute of Technology with a CGPA of 9.29 / 10 , recognizing consistent academic excellence.
Recipient of the KITE Achiever Award and selected for national-level hackathons including Smart India Hackathon (SIH) , Trixathon 2K25 , and CITHackathon .
Ranked in the Top ~3% globally on Kaggle and achieved Rank 72 out of 2,584 participants in the <i>Spaceship Titanic</i> machine learning competition.
Secured a Top 24 position out of 220+ teams at the CIT Hackathon 2025 (12-Hour National Hackathon) for <i>MedCycle</i> , an AI-powered medicine redistribution web application; awarded Certificate and Bronze Medal

Certifications

• OCI 2025 Certified AI Foundations Associate • Introduction to Data Analysis using Excel • Neural Networks and Deep Learning	• IBM Data Science Professional Certificate • OCI 2025 Certified Generative AI Professional • Machine Learning with Python
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